

DATA SCIENCE ETHICS



The **Data Science Ethics** course has significantly reshaped the way I understand the responsibilities involved in working with data. Before taking this course, I viewed ethics as a theoretical or optional layer added to technical work. However, the materials, case studies, and examples in the course demonstrated that ethical thinking is not optional at all—it is essential for protecting individuals, avoiding harm, and ensuring that data science contributes positively to society. This reflection discusses the ethical concepts that stood out to me, how the course influenced my views, the lessons I will carry forward, and the real-world situations I can now see differently because of what I learned.

Ethical Concepts and Case Studies that Stood Out

One of the most impactful concepts in the course was the idea of **simple utilitarian ethics**. The course emphasized that in order for data work to benefit society, there must be a shared understanding of right and wrong. Instead of relying on complex philosophical theories, the course introduced a practical framework focused on minimizing harm and maximizing benefit. This “common ethical ground” created a foundation for all the examples and discussions that followed.

The course's discussion of **data ownership** also left a strong impression on me. The question “If I take a photo of you, who owns it—you or me?” served as a powerful analogy for data ownership. The course highlighted that “data about me is often not owned by me,” which made me realize how easily individuals can lose control over their personal information once it is collected.

Among the case studies, the **Tuskegee Syphilis Study** was the most disturbing. The course provided detailed explanation of how 600 men were misled about their condition and denied treatment even after penicillin became widely available. This extreme violation of trust and dignity showed why **informed consent**, voluntary participation, and the ability to withdraw are non-negotiable principles. This case directly led to the establishment of the **Institutional Review Board (IRB)**, which still protects human subjects today.

More modern examples, such as the **Facebook Emotional Contagion experiment** and **OkCupid's manipulation of compatibility scores**, helped me see how unethical practices still occur in the digital age. Even though these companies did not physically harm users, the experiments violated expectations of transparency. The course made it clear that the difference between acceptable A/B testing and unethical research lies in **whether participants know they are being experimented on**.

How the Course Changed or Reinforced My Views

This course fundamentally changed how I see privacy. The analogy of the **Panopticon**, where prisoners behave as if they are constantly watched, helped me understand how surveillance—even passive data collection—can influence human behavior. Privacy is not just about hiding secrets; it is about preserving autonomy.

The course also corrected my assumption that legality equals ethical correctness. Many actions may be legal, such as collecting metadata or selling customer information during bankruptcy, yet still be unethical because users never expected their data to be used in those ways. This distinction pushed me to think more critically about fairness, transparency, and the social consequences of data decisions.

Lessons I Will Carry Forward

Several lessons from this course will stay with me in my academic and professional life:

1. **The six-word code: "Don't surprise people. Own the outcomes."**
This principle captures the core of ethical data practice—be transparent and take responsibility.
2. **Informed consent must be meaningful.**
It is not enough to show users a long policy; they must genuinely understand what will happen to their data.
3. **Data persists beyond the company that collected it.**
The example of RadioShack selling user data after bankruptcy taught me that data handling must consider long-term consequences.
4. **De-anonymization is a real risk.**
Case examples such as the AOL and Netflix datasets showed how easily "anonymous" data can be re-identified.

Real-World Applications

After this course, I now see many everyday situations differently. Mobile apps requesting unnecessary permissions resemble the "waste data collection" described in the course. Location data that seems harmless can actually reveal sensitive information, such as religious habits or home routines. Social media algorithms can influence emotions, just like the Facebook study demonstrated.

Conclusion

Overall, the **Data Science Ethics** course gave me a strong ethical foundation that I will rely on throughout my career. It helped me understand not only what ethical principles are, but why they matter and how to apply them in real situations. I now feel more confident in my ability to identify risks, protect user privacy, and use data responsibly. This course has prepared me to be not only a skilled data practitioner, but an ethical one.